

Language-Conditioned Rank Reversal in Agentic LLM Judges

Anonymous Author(s)

Submission Id: 1655

Abstract

Agent-based evaluators assess artifacts produced by autonomous developer agents and multiagent teams. Their scores can change when written requirements and judge instructions are localized, even if the code is identical. This makes evaluator selection and quality-gate decisions sensitive to the judging language. We present consensus-based calibration (CBC), a label-free method that estimates and removes the language-by-evaluator interaction while preserving each language’s evaluator-panel mean. CBC applies standard two-way double centering to a balanced score panel; we establish identification conditions and a task-level uncertainty bound. We evaluate it on 55 DevAI tasks with code held fixed across eight languages: English, Arabic, Turkish, Chinese, Hindi, Japanese, Spanish, and Swahili. Held-out analyses show more consistent evaluator severity rankings and quality-gate decisions across languages, with similar ranking results on a separate preference benchmark. In a task subset, the largest interaction exceeds average repeat-call variation. The correction targets consistency; it leaves shared language shifts unchanged and does not establish improved judging accuracy. Localized inputs still combine judge behavior with translation effects. Code is available at <https://anonymous.4open.science/r/multilingual-judge-calibration-2663/>.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; **Multi-agent systems**; • **General and reference** → *Evaluation*.

Keywords

Agent-as-a-Judge, multilingual evaluation, LLM evaluation, calibration, language bias

ACM Reference Format:

Anonymous Author(s). 2027. Language-Conditioned Rank Reversal in Agentic LLM Judges. In *Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027)*, Hanoi, Vietnam, 3–7 May 2027, IFAAMAS, 8 pages.

1 Introduction

Agentic systems increasingly use language models to assess work produced by developer agents, including multiagent teams. Agent-as-a-Judge uses file-location, reading, and evidence-retrieval modules to assess artifacts against written requirements [28]. Its scores can decide whether a task passes a quality gate. The reliability of this judging agent therefore affects both agent-performance measurement and artifact acceptance. If the score changes with the

language of the judging inputs, the same artifact may receive a different decision.

Multilingual judge studies report inconsistent scores and preferences [7, 9, 24]. In many studies, changing the language also changes the response being judged. A score difference may then come from the judge or from the translated response. Section 3 explains why scores alone cannot separate these explanations. Judge-panel averaging and psychometric audits [22, 27] address other sources of evaluator variation. Our target is an offset that varies jointly with language and evaluator.

We hold the code artifact fixed and vary the localized judging inputs: requirements, rubric, and judge instructions. This design attributes the observed score differences to the judging setup rather than to changed code. It does not separate the model’s behavior from differences introduced when the requirements are translated. We estimate both a shift shared across evaluators and an interaction between language and evaluator. We then ask whether that interaction changes quality-gate decisions and whether it is larger than variation between repeat judge calls.

We make three contributions. First, a fixed-artifact audit shows that evaluator severity can reverse across languages even when the judged code is byte-identical, with direct consequences for quality-gate decisions. Second, consensus-based calibration (CBC) estimates and removes the language-by-evaluator interaction without human labels; we give conditions for its identification and concentration. Third, held-out analyses on DevAI and a separate preference panel test whether the estimated interaction persists beyond the fitting tasks, while repeat-call experiments distinguish its largest effects from ordinary judge variability. We also state explicitly what the available human labels can and cannot establish about judging accuracy. Figure 1 summarizes the study design and analysis.

2 Related Work

LLM-judge reliability. Zheng et al. [26] document position, verbosity, and self-enhancement biases when LLMs judge conversational responses. Across twenty NLP tasks, Bavaresco et al. [1] find that agreement with human judgments varies with the model, task, and property being evaluated. Ratings can also vary between calls within one language [21]. Psychometric audits measure how prompts change decision criteria and sensitivity to controlled quality changes [22]. These findings motivate testing both systematic language effects and repeat-call variation, rather than treating every score difference as a stable evaluator characteristic.

Multilingual evaluation. Hada et al. [9] compare multilingual ratings with native-speaker judgments and find score inflation. Other studies report differences in judge consistency and reliability across languages [7, 12]. MM-Eval assesses multilingual judging through both pairwise decisions and the consistency and fairness of absolute scores [20]. The CIA suite combines a human-annotated multilingual test set with Hercule, an evaluator that uses English reference answers to score responses in other languages [4]. Sheth et al. [19]

This work is licensed under a Creative Commons Attribution International 4.0 License.

Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027), M. Baldoni, F. Fang, W. Yeoh, N. Yorke-Smith (eds.), 3–7 May 2027, Hanoi, Vietnam. © 2027 International Foundation for Autonomous Agents and Multiagent Systems (www.ifaamas.org).

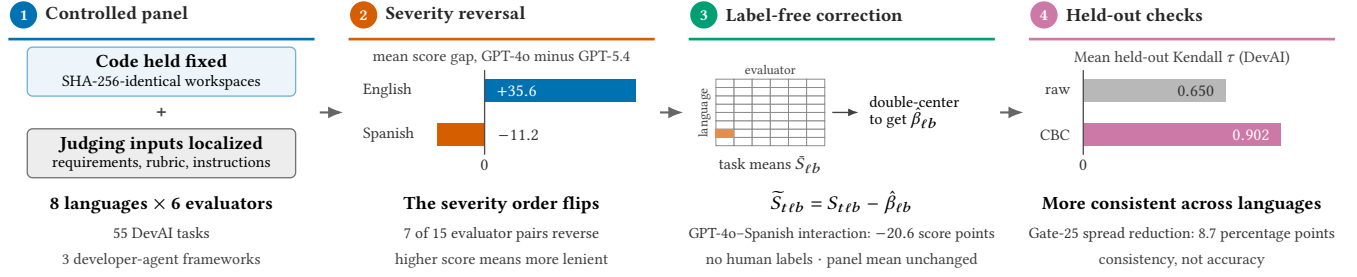


Figure 1: Study overview. (1) Code is held fixed while the judging inputs are localized. (2) GPT-4o scores above GPT-5.4 in English and below it in Spanish (Table 1). (3) CBC double-centers the task-mean score matrix to estimate and remove the language-by-evaluator interaction. (4) On held-out tasks, severity rankings and gates agree more across languages (Table 2, Sec. 5); this is consistency, not accuracy. The grid is schematic.

decompose evaluations into shared criteria and learn a transfer module from English labels, without target-language annotations. These approaches evaluate or adapt multilingual judges; we estimate interactions from an already collected score panel. Yin [24] test preference stability under English, Chinese, and mixed-language presentations of response pairs, including translated responses. Our fixed-code design instead changes the judging inputs and measures quality-gate consequences. It still mixes judge behavior with translation of the requirements.

Calibration and aggregation. CalibraEval uses label-free distribution correction to reduce selection bias when response positions or option identifiers change [14]. Dawid–Skene estimates observer error rates for categorical labels when the true labels are unavailable [3]. Such latent-label estimation has a different target from decomposing absolute score means. Judge-aware pairwise models estimate evaluator-specific discrimination [23], while heterogeneous judge-aware ranking models consensus and structured disagreement [25]. Round-robin assignment can recover a benchmark’s judge-panel mean at a single-judge budget [27]. Labeled calibration uses signals from a diverse judge panel [15]; human anchors are needed to test accuracy [13]. Our CBC applies standard two-way ANOVA double centering [17, 18] to a language-by-evaluator interaction. It requires a balanced panel for the stated identification result and leaves the shared language shift intact.

Agentic code judging. Agent-as-a-Judge provides requirement-level evaluation of agent-produced artifacts and the DevAI benchmark [28]. An earlier five-language localization study established that judge scores vary with language and evaluator backbone in this setting [16]. The current study addresses a different question: whether the language-by-evaluator interaction can be identified from a balanced judge panel and whether removing it stabilizes evaluator severity rankings and downstream quality gates. The analysis covers eight languages, derives formal properties of the interaction estimate, and evaluates the correction on held-out tasks. Automated validation of software-artifact evaluators provides a complementary approach to testing judge reliability [5]. A separate M-RewardBench preference panel [8] tests whether the ranking-consistency pattern also appears under a different benchmark and collection protocol. Neither analysis establishes that the corrected judgments are more accurate.

3 Score Model and Fixed Artifacts

Let t , ℓ , and b index tasks, languages, and evaluator backbones. Let n , k , and m be their respective counts. The score is the mean across the three balanced developer-agent framework arms for that cell. We use the following additive model:

$$S(t, \ell, b) = \mu(t) + \alpha(b) + g(\ell) + \beta(\ell, b) + \gamma(t, \ell) + \epsilon(t, \ell, b), \quad (1)$$

where μ is a task effect, α an evaluator offset (severity), g a language shift shared by evaluators, β the language-by-evaluator interaction, and γ a task-language shift shared by evaluators. We normalize β to have zero row and column sums and γ to have zero task mean in each language. For the theoretical results, we assume $\mathbb{E}[\epsilon(t, \ell, b)] = 0$ in every cell. Framework effects that do not interact with both language and evaluator cancel under double centering in this balanced panel. Task-specific language-by-evaluator deviations enter ϵ ; the original panel cannot separate them from repeat-call noise. Thus β is the task-average interaction for this panel. These terms describe scores, not judging accuracy.

If the evaluated response is also translated, the same score table can arise from a language-invariant evaluator seeing different responses or a language-sensitive evaluator seeing equivalent responses. We avoid that particular ambiguity by judging the same code. An audit of the six evaluator packs, three framework directories, and their workspace trees found the same 4,878-file SHA-256 manifest in all eight conditions. The localized rubric and requirements may still differ in meaning, so β describes the judging setup as a whole.

4 A Fixed-Artifact Panel

We extend the earlier five-language Agent-as-a-Judge setting [16] with Japanese, Spanish, and Swahili. The panel has 7,920 judge runs over 6 evaluators, 8 languages, 3 developer-agent frameworks, and 55 DevAI tasks [28], a complete balanced panel of $55 \times 8 \times 6$ task-level cells, each the mean of three framework runs. The languages are English, Arabic, Turkish, Chinese, Hindi, Japanese, Spanish, and Swahili. The framework arms are MetaGPT, GPT-Pilot, and OpenHands. MetaGPT supplies artifacts from collaborating agents with specialized development roles [10]. Each evaluator backbone runs as a separate judging condition. The six evaluator backbones are GPT-4o, GPT-5.4, Claude Sonnet 4.6, Gemini 3 Flash Preview,

DeepSeek-V3.2, and Qwen3.5-9B. Figures 2 and 3 shorten Claude Sonnet 4.6 to “Sonnet 4.6” and Gemini 3 Flash Preview to “Gemini 3 Flash” to fit their axes. The rerun logs record exact provider-qualified routes; the original panel did not preserve model snapshot dates. For the replication reruns, we record the model route, temperature, top- p , workflow, and manifest seed (the seed is not passed to the provider) in the accompanying JSONL logs. The judge uses the released parser: a response containing <SATISFIED> is satisfied, and every other response, including an empty or failed response, is unsatisfied. The task score is 100 times the unweighted fraction of requirements marked satisfied; recorded prerequisites are not used in that aggregate. A task passes a gate if the score strictly exceeds the threshold. Requirement text and user queries were translated from English by GPT-4o under a token-preservation instruction (paths, identifiers, and backtick spans held fixed); judge templates were localized while keeping the English <SATISFIED>/<UNSATISFIED> tags. Native speakers with graduate training reviewed all eight language versions for fluency and technical fidelity, following the protocol of the five-language setting [16]. Localized system prompts additionally have mean LaBSE cosine similarity 0.90 with English (range 0.83–0.95 over 35 prompt–language pairs) [6]. A structural audit of the 55-instance MetaGPT pack finds no requirement-count mismatches, backtick-span mismatches in 14 of 385 non-English instances, and complete tag preservation in every judge prompt. The released original-panel artifacts do not preserve provider snapshot dates, base URLs, or per-call retry logs; we therefore reproduce those results from the saved score matrices rather than claim a full provider-level regeneration.

Evaluators differ substantially in mean scores: mean satisfaction awarded runs from 42.9 (Gemini 3 Flash Preview) to 3.6 (Qwen3.5-9B), and Qwen3.5-9B awards zero on 82.0% of cells. These means order evaluator *leniency*, not evaluator quality.

4.1 Consensus-based calibration

Let $\bar{S}_{\ell b}$ be the task mean, with a dot denoting an average over that index. We estimate the interaction and adjust scores by

$$\hat{\beta}_{\ell b} = \bar{S}_{\ell b} - \bar{S}_{\cdot b} - \bar{S}_{\ell \cdot} + \bar{S}_{\cdot \cdot}, \quad \tilde{S}_{\ell b} = S_{\ell b} - \hat{\beta}_{\ell b}. \quad (2)$$

We call this consensus-based calibration (CBC). Here, calibration refers to removing panel-relative language-by-evaluator offsets, not calibration to human correctness. Double centering is a standard two-way ANOVA operation, not a new estimator [17, 18]. Its target is relative to the evaluator pool and benchmark. Adjusted scores are not clipped to [0, 100]; clipping would break panel-mean invariance.

PROPOSITION 1 (IDENTIFICATION AND PANEL-MEAN INVARIANCE). *In a complete balanced panel under Eq. 1, $\mathbb{E}[\hat{\beta}_{\ell b}] = \beta_{\ell b}$ for any task-language effect $\gamma(t, \ell)$ shared by evaluators. Moreover, $m^{-1} \sum_b \tilde{S}_{\ell b} = m^{-1} \sum_b S(t, \ell, b)$ for every task and language.*

Proof. Task averaging makes μ a common constant and removes the normalized γ . The four terms in Eq. 2 cancel that constant, α , and g , leaving normalized β and mean-zero error. Thus $\mathbb{E}[\hat{\beta}_{\ell b}] = \beta_{\ell b}$. The second identity follows from $\sum_b \hat{\beta}_{\ell b} = 0$: subtracting the interaction preserves each task-language panel mean. $\square$

The identity means CBC cannot change a decision based on the mean score of all evaluators for one item. It can change an

individual evaluator’s decision or the order of evaluator severity. It leaves the shift g shared by evaluators untouched.

PROPOSITION 2 (TASK-LEVEL UNCERTAINTY). *Assume $k, m \geq 2$, tasks are independent, scores lie in $[0, 100]$, and the double-centered task error has mean zero. Dependence across languages and evaluators within a task is allowed. With $c = (1 - 1/k)(1 - 1/m)$, for every $x > 0$,*

$$\Pr\left\{|\hat{\beta}_{\ell b} - \beta_{\ell b}| \geq x\right\} \leq 2 \exp\left(-\frac{nx^2}{80000c^2}\right).$$

For all mk cells at once, multiply the right side by mk .

Proof. Double centering one task gives a linear combination whose positive coefficients sum to $2c$, as do the absolute negative coefficients. It lies in $[-200c, 200c]$. Hoeffding’s inequality across independent tasks and a union bound give the result. $\square$

This conservative bound establishes concentration toward the model target as independent tasks increase. Independent language-evaluator cell noise would give a tighter bound, but judging the same artifact in every condition makes that assumption implausible. We use a task bootstrap for empirical intervals. For gate analyses, we fit $\hat{\beta}$ on 1,000 bootstrap samples and evaluate on omitted tasks from the same observed language set.

For two evaluators i, j , their language-specific mean score gap is

$$d_{ij}(\ell) = \bar{S}_{\ell i} - \bar{S}_{\ell j} = \alpha_i - \alpha_j + \beta_{\ell i} - \beta_{\ell j} + \bar{\epsilon}_{\ell i} - \bar{\epsilon}_{\ell j}. \quad (3)$$

The task, shared-language, and task-language terms cancel. We call a pair a *severity reversal* when observed gaps have opposite signs in two languages. This rules out one observed ordering of severity that fits every language. It does not rank evaluator quality: higher scores can simply mean greater leniency.

5 Results

Localizing the judging inputs changes evaluators’ relative severity. We first test that result, then examine its effect on threshold decisions.

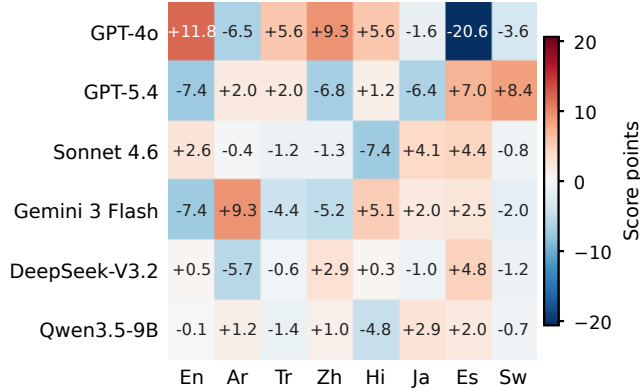
5.1 Language reverses which evaluator is more severe

Seven of fifteen evaluator pairs reverse their average severity order across languages (Table 1). The table’s witness languages are selected from 28 possible pairs and are descriptive, so we test the total reversal count. We fit an additive task, language, and evaluator null, permute its residuals over languages independently within each (task, evaluator) group, and add the fitted values back on 1,000 draws. This diagnostic assumes residual exchangeability across languages, a stronger condition than Proposition 2. The null has a median of one reversal and a maximum of three, against seven observed (permutation $p = 0.001$). Figure 2 shows the interaction: GPT-4o is -20.6 points in Spanish and $+11.8$ in English, relative to its own mean and the shared language effect. Pointwise 95% bootstrap intervals exclude zero for 34 of 48 cells. The permutation test applies to the total reversal count, not to each selected witness pair. The cell intervals are not multiplicity-adjusted; gate analyses are exploratory.

Table 1: Seven evaluator pairs reverse severity order. Gaps are mean score differences for the first evaluator.

Pair	Gaps (pts)	
GPT-4o vs. GPT-5.4	En +35.6	Es −11.2
GPT-4o vs. Gemini 3 Flash Preview	En +9.5	Es −32.8
GPT-4o vs. Claude Sonnet 4.6	Hi +30.7	Es −7.2
GPT-4o vs. DeepSeek-V3.2	En +32.4	Es −4.3
GPT-5.4 vs. Claude Sonnet 4.6	Ja −9.0	Sw +10.7
GPT-5.4 vs. DeepSeek-V3.2	Zh −5.0	Sw +14.3
Claude Sonnet 4.6 vs. DeepSeek-V3.2	Ar +8.5	Hi −4.5

En = English; Es = Spanish; Hi = Hindi; Ja = Japanese; Sw = Swahili; Zh = Chinese; Ar = Arabic; Tr = Turkish. Each row uses its most opposed language gaps.

**Figure 2: Centered language-by-evaluator interaction (score points) on fixed code. Positive values mean higher scores. Language abbreviations are defined under Table 1; evaluator short labels are defined in §4.**

5.2 Held-out ranking consistency

We evaluate CBC on tasks omitted from each of 1,000 bootstrap fitting samples, while all eight languages remain observed. The reported percentile intervals are computed across these task-bootstrap fits and their corresponding omitted-task evaluations. The metric is mean pairwise Kendall τ between language-specific evaluator *severity* rankings, over all 28 language pairs. Cross-language agreement is a target-aligned diagnostic: CBC is designed to remove stable language-by-evaluator offsets. The held-out analysis tests whether those offsets persist across omitted tasks rather than only describing the fitting sample; it does not establish that the removed variation is erroneous. On DevAI, it rises from 0.650 to 0.902 (Table 2). ComBat was developed for microarray batch correction using empirical-Bayes location and scale estimates [11]. Our ComBat-inspired comparator treats language as a batch, shrinks one residual location and scale per language toward zero and one, and restores evaluator main effects. It uses simpler pooling than standard ComBat and does not fit language-by-evaluator interactions. Quantile normalization [2] maps each evaluator column’s ordered language means to a common distribution fitted across evaluator columns; ties receive averaged target values. Both comparators fit on in-bag tasks and apply the fitted adjustments to omitted-task means.

Table 2: Held-out cross-language ranking agreement. Values are mean Kendall τ with 95% percentile bootstrap intervals. Bold marks the highest mean per panel.

Panel	Method	τ	95% interval
DevAI	Raw	0.650	[0.610, 0.714]
	ComBat-inspired	0.650	[0.610, 0.714]
	Quantile	−0.097	[−0.119, −0.068]
	CBC	0.902	[0.795, 0.968]
M-RewardBench	Raw	0.430	[0.371, 0.486]
	Adapted BTL	0.405	[0.352, 0.467]
	CBC	0.900	[0.790, 1.000]

CBC instead subtracts interaction offsets while retaining language and evaluator main effects. The ComBat-inspired adjustment gives nearly the same agreement as raw scoring (equal at three decimals), and this quantile adaptation performs worse. The paired CBC minus ComBat-inspired difference is positive in all 1,000 bootstrap replicates. These rankings measure which evaluator awards higher scores, not which one judges code more accurately.

On a separate M-RewardBench preference panel [8], we collected pointwise 1–5 scores for the chosen and rejected response of the first 1,500 dataset IDs (805 alpaca-eval-easy, 695 alpaca-eval-hard), aligned across English, Arabic, Turkish, Chinese, Hindi, Japanese, and Spanish, using GPT-4o, GPT-5.4, Claude Sonnet 4.6, Gemini 3 Flash Preview, and DeepSeek-V3.2. The resulting 10,500 language-item instances define 105,000 pointwise score slots. Thirty-five of the 52,500 evaluator-item-language margins are missing from the saved score table; the analysis averages available margins within each cell, excluding missing scores from weighted denominators. CBC acts on chosen-minus-rejected score margins, which measure preference strength rather than evaluator severity. Under the same bootstrap fit and omitted-item evaluation, mean cross-language τ rises from 0.430 to 0.900 (Table 2). An adapted judge-aware Bradley–Terry–Luce (BTL) comparator [23] obtains 0.405. The cited model fits latent rankings with judge-specific discrimination parameters. Our adaptation converts differences between evaluator preference margins into pairwise wins, fits evaluator rankings separately per language, and uses the easy/hard dataset strata for discrimination parameters. It fits directly on omitted-item comparisons, excluding missing margins and assigning exact ties half a win. CBC learns continuous-margin interactions on in-bag items. Thus this adaptation compares ranking procedures with different targets and fitting data; it does not establish superiority over the original judge-aware calibration framework. The instance source and score-collection protocol differ from DevAI, so this result shows ranking consistency on another panel, not fixed-code attribution.

On the full fitting panels, CBC yields $\tau = 1$ by construction: the corrected evaluator means differ between languages only by a shared additive shift, so their severity or preference-strength rankings coincide. The omitted-item results in Table 2 are the relevant stability checks. A leave-one-language-out DevAI diagnostic rises only from 0.649 to 0.740 when the unseen language receives zero estimated interaction. It does not establish transfer to unseen languages.

Table 3: DevAI evaluator and task-count ablations, with 100 bootstrap replicates per evaluator subset or task count. Tasks are sampled uniformly. Beta difference is mean absolute difference from the full-panel estimate.

Evaluators m	2	3	4	5	6
Mean τ	0.901	0.907	0.900	0.898	0.911
Across-subset sd	0.206	0.105	0.075	0.044	–
Tasks n	10	20	30	40	55
Mean τ	0.759	0.838	0.853	0.891	0.906
Replicate sd	0.103	0.061	0.057	0.054	0.051
Mean beta diff. (pt)	2.01	1.43	1.11	0.91	0.67

5.3 Model checks and ablations

We fit Eq. 1 to all 7,920 framework-level observations with framework fixed effects. A simplified version without $\gamma(t, \ell)$ has $R^2 = 0.691$; including task-language terms gives $R^2 = 0.705$. This modest in-sample increase of 0.015 is descriptive; adding terms improves training fit by construction. Unexplained variation warns against treating the interaction as a complete causal account.

Table 3 shows two checks from the DevAI data. We enumerate every evaluator subset at each pool size, using 100 task-bootstrap replicates per subset. Mean held-out τ stays near 0.90, but two-evaluator subsets vary much more than larger subsets. For each task count $n \in \{10, 20, 30, 40, 55\}$, we run 100 replicates with RNG seed 107. Each replicate selects n of the 55 tasks uniformly without replacement, then bootstraps n draws within that subset. CBC fits on those draws and evaluates omitted tasks from the same subset. Selection is not stratified by task category; each task has equal selection probability, so category coverage can vary between replicates. It shows the estimate approaching the full-panel interaction as the number of tasks grows. The mean absolute difference falls from 2.01 points with 10 tasks to 0.67 with 55 tasks (Table 3); the full-panel interaction is a reference estimate, not ground truth. The $n = 55$ value in that experiment differs slightly from the 1,000-replicate headline because the two loops use independent seeds and protocols. Splitting requirements by type also gives a larger consistency gain for model-construction and evaluation-metric checks (0.737 to 0.840) than for data-loading and training checks (0.770 to 0.789). These are descriptive ablations, not evidence that semantic requirements have a uniquely causal language effect.

We test panel incompleteness using the saved scores. In 500 simulations per rate, we independently remove task–language–evaluator cells with probability 0.05 or 0.10. Relative to the complete-panel estimate, mean absolute interaction error is 0.34 or 0.51 points using available cell means, and 0.23 or 0.33 after fitting task and cell fixed effects. This tests random missingness only; it does not extend Proposition 1 to systematic gaps. Task-specific centered interactions vary, with a root-mean-square deviation of 6.5 points from the pooled estimate. The GPT-4o Spanish interaction is negative for all 55 tasks, but these residuals mix judge noise with genuine task variation.

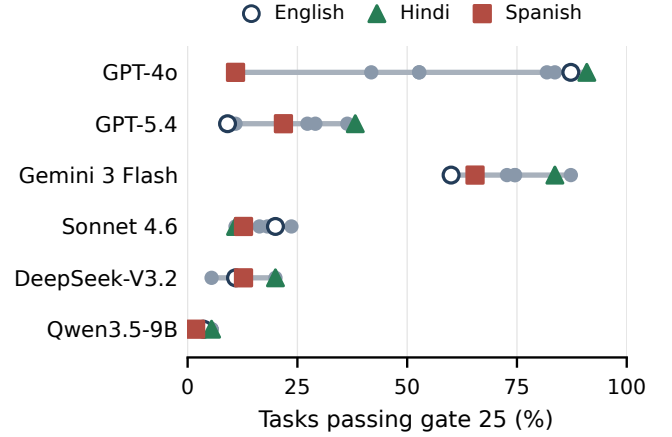


Figure 3: Pass rates at gate 25 for the same 55 code tasks. Lines span all eight languages. Symbols mark the English reference and GPT-4o’s observed extremes, Hindi and Spanish; other languages are gray dots. Evaluator short labels follow §4.

5.4 Framework comparisons are descriptive

The three developer-agent frameworks have close scores. Their largest mean gap within a language is 1.50 points. These descriptive means do not establish a framework ranking that generalizes across tasks and evaluators. The descriptive evaluator-by-framework interaction has a largest cell of 1.76 points; the additive model is an approximation. Gate decisions depend directly on score levels, which we examine next.

5.5 Gate pass rates vary by 80 percentage points

For a fixed evaluator on identical code, the fraction of tasks passing a satisfaction gate depends heavily on the judging language: at a gate of 25, GPT-4o passes 90.9% in Hindi and 10.9% in Spanish (Fig. 3). We highlight these languages because they give GPT-4o’s highest and lowest observed gate-25 pass rates; English is the reference. They are descriptive examples, not preselected language groups or evidence of a language-wide cause. All eight languages enter the interaction estimates, reversal search, and consistency metrics. Sweeping the threshold, the correction cuts across-language spread by 10.6 to 14.2 percentage points at the tested gates from 10 to 30, where scores concentrate (Fig. 4). The changes are smaller and can worsen spread at higher gates, where fewer tasks pass. For each evaluator, spread is the maximum minus minimum pass rate across languages; we report its mean over evaluators.

The direction of the shift differs by evaluator. Comparing the English default against the mean of the seven localized conditions at gate 25, GPT-4o becomes stricter when the rubric is localized (−27.9 percentage points, 95% CI [−37.7, −18.2]), while Gemini 3 Flash Preview and GPT-5.4 become more lenient (+13.9 [+7.0, +21.3] and +15.6 [+9.4, +22.3]). The remaining three evaluators show contrasts whose intervals include zero. Localizing an evaluation harness can tighten or loosen the same gate, with the direction depending on the evaluator. A single language adjustment cannot remove shifts that differ in sign across evaluators.

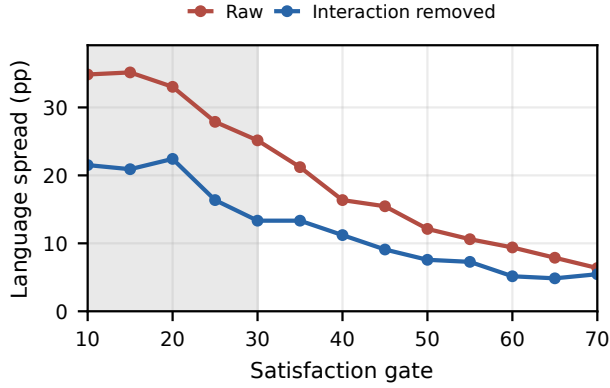


Figure 4: Mean pass-rate spread across languages as the gate changes. The shaded range marks gates 10–30. Curves use the saved score panel.

Table 4: Mean language spread in gate pass rates (percentage points) on the full panel. Bold marks the lowest spread at each gate.

Correction	Gate 10	Gate 25	Gate 50
None (raw)	34.8	27.9	12.1
Per-language centering	39.4	28.5	12.7
CBC ($\hat{\beta}$)	21.5	16.4	7.6

A shift shared by all evaluators survives double centering. Its estimated range is 8.6 points across the eight languages, from Hindi +5.3 to Spanish −3.3. It can be centered, but human labels are needed to judge whether that improves accuracy.

5.6 Comparison with simpler corrections

The gate endpoint can change under any score transformation. Table 4 compares the interaction correction with centering each language by its overall mean. The interaction correction gives the lowest spread at each shown gate. In the full panel, gate-25 spread falls from 27.9 to 16.4 percentage points. When we fit on bootstrap samples and evaluate on omitted tasks, the mean reduction is 8.7 percentage points (95% percentile interval [3.8, 14.0]; 998 of 1,000 replicates are positive). This is evidence of more consistent gates, not more accurate judgments.

5.7 Robustness

Leaving out each task in turn moves $\max |\hat{\beta}|$ within [20.2, 20.9], and no single task moves it by more than 0.4 points. Table 5 shows that removing GPT-4o reduces reversals from 7/15 to 3/10 pairs and the largest interaction from 20.6 to 8.0 points. Removing any other evaluator leaves the largest interaction at 19.2 points or more. Qwen3.5-9B is a separate concern: it awards zero on 82.0% of cells and has scores concentrated at zero, so we also check the panel without it. Of 52,560 requirement judgments, 4.6% lack a parseable

tag or are empty, and 2,301 of those 2,410 cases are empty Qwen3.5-9B responses: 62% of its Spanish requirements and 67% of its Swahili ones. The released parser scores those empty strings unsatisfied. Excluding Qwen3.5-9B, the noncompliance rate is 0.12%, and GPT-4o’s Spanish missing-tag rate is 0.5%, so missing tags do not explain the 20.6-point cell. We retain Qwen3.5-9B because parser-visible failures are part of the released protocol’s behavior, and report reduced-panel analyses to check the effect of this zero-score concentration.

Because four of the seven reversals involve GPT-4o, we repeat the analysis without it. Among the five remaining evaluators, the largest interaction is Gemini 3 Flash Preview in Arabic at +8.0 points (95% CI [6.3, 9.9]), and gate spread still falls from 27.6 to 21.5 points at a gate of 10 and from 17.5 to 13.8 at 25. The largest interaction is roughly 39% of its full-panel size.

Because both the requirement text and the judge instructions are localized, $\hat{\beta}$ still mixes judge behavior with specification-translation quality. A control that keeps benchmark text localized and varies judge-instruction localization was previously run at 20-task, two-language, one-evaluator scale [16] and found a large Hindi/Arabic asymmetry. That control is not extended to this panel, so it cannot separate these sources across our eight languages and six evaluators.

An additive model on [0, 100] can strain linearity at the floor. Recovering $\hat{\beta}$ after an arcsin-square-root transform of the satisfaction proportion agrees with the linear matrix (Spearman 0.97; no sign flips among cells with $|\hat{\beta}| \geq 1$), and the gate-25 spread reduction remains 11.5 points when corrected angles are clipped to $[0, \pi/2]$ before mapping back to scores. A percentile-rank transform agrees similarly (Spearman 0.95). A logit link does not: it flips five of those cells and widens gate spread, because Qwen3.5-9B’s pile-up at zero is mapped to large logit residuals. We keep the linear correction for the gate endpoint; the logit sensitivity check shows that conclusions need not survive every transformation.

Double centering removes the variation that cross-language consistency measures penalize. A higher consistency score therefore cannot show that the removed term was error. The next section tests repeat-run variation; §7 explains why the available human labels cannot support the planned requirement-level accuracy test with the saved judgments.

6 Replication

The released panel holds one judgment per (evaluator, language, framework, task) cell. It therefore cannot separate repeat-call variation from a stable language effect. We measure that variation with new calls.

We reran a stratified subset chosen before any call was made: 10 tasks spread across the requirement-count distribution, all eight languages, all six evaluators, three repeats, and the MetaGPT arm only (1,440 calls). The reruns reuse the released judging workflow and parser; text judging calls request temperature 0 and top- p 0.9. The manifest seed is recorded and is not sent to the provider. The decision rule was fixed before the reruns: retain the magnitude claim only if the test–retest standard deviation s , averaged over tasks, languages, and evaluators, is below one quarter of the largest interaction B . This practical criterion requires the largest interaction

Table 5: Evaluator behavior and leave-one-evaluator-out sensitivity. Mean score and zero rate use each evaluator’s 440 task-level cells. Interaction magnitudes are score points. Bold identifies GPT-4o’s effect and Qwen3.5-9B’s zero-score concentration, not evaluator quality.

Evaluator	Full panel			After removing this evaluator	
	Mean score	Zero cells (%)	Own max $ \hat{\beta} $	Reversing pairs	Panel max $ \hat{\beta} $
Gemini 3 Flash Preview	42.9	2.7	9.3	6/10	20.1
GPT-4o	33.2	6.1	20.6	3/10	8.0
GPT-5.4	16.8	22.7	8.4	4/10	19.2
Claude Sonnet 4.6	15.4	23.4	7.4	4/10	19.7
DeepSeek-V3.2	12.2	33.0	5.7	4/10	19.7
Qwen3.5-9B	3.6	82.0	4.8	7/10	20.2

to exceed average repeat-call variation by a factor of four. Otherwise the claim is qualified or withdrawn as specified in the analysis script.

The 1,440 calls completed. We compute a sample standard deviation across the three repeats for each task–language–evaluator cell, then average those standard deviations. Mean test–retest sd is $s = 4.08$ points against $B = 20.61$, a ratio of 0.20, so the prespecified rule passes. The largest mean sd for one language-by-evaluator cell is 10.9 points; individual tasks can be noisier. This is a 10-task MetaGPT slice, not a repeat of the full 55-task, three-framework panel. Across languages, mean cell sd ranges from 3.09 points in English to 5.06 in Spanish; this slice cannot establish a language-resource trend.

7 Is the Removed Term Error?

Exceeding repeat-call variation does not prove the interaction is error: it may encode language-dependent judging competence. DevAI supplies expert satisfaction labels [28]; the local export has 1,098 judgments over 55 tasks and three agents. Only 36.5% are satisfied, so always predicting unsatisfied gives 63.5% accuracy. A human-label test should compare paired requirement decisions using balanced accuracy and Matthews correlation, retaining requirements and frameworks together in task resampling. Matching aggregate human satisfaction rates cannot establish correctness. Since requirements and human labels are in English, results should be reported with and without English.

This submission cannot run that test. The export has five human-label requirements and four evaluator requirements for 52_DevIn_AI_Trains_an_AI in each agent workspace, and the saved judgments contain binary verdicts without a continuous requirement-level score from which to compute corrected verdicts. Reporting a pooled balanced-accuracy difference would require reindexing a requirement or inventing scores. The artifact records this blocked test. Comparing task scores with aggregated human satisfaction would address a coarser, currently untested target.

M-RewardBench supplies gold preferences. A prior analysis reported a panel-mean agreement gain from 68.7% to 76.6% after CBC. Auditing its 700 sampled item-language margins identifies a numerical tie artifact: 55 apparent flips have zero raw margins and tiny positive corrected residuals. Three incomplete-cell margins differ slightly without changing decisions. Using the same absolute-margin tie tolerance of 10^{-9} , agreement is 68.7% before

and after CBC. Proposition 1 gives panel-mean invariance for complete panels. We retract the earlier accuracy gain and make no human-accuracy claim for CBC.

Limitations

The evidence covers 55 tasks from one code-evaluation family and six evaluator routes. Repeat calls cover only 10 MetaGPT tasks, with some task-level standard deviations far above the pooled mean; the largest interaction depends on GPT-4o. The external panel covers two AlpacaEval subsets and has 35 missing margins, so complete-panel identification is not exact for its available-case analysis. Without an instruction-only control at this scale, $\hat{\beta}$ combines judge behavior with translation quality. The human-label test is blocked by a requirement mismatch and binary-only judgments; this is not evidence of a null accuracy effect. CBC leaves shared language shifts intact, and human labels are needed to assess whether removing them improves accuracy. The additive model also strains near the score floor: arcsin-square-root checks agree with the linear results, but the logit check does not. Corrected scores are not ground truth. Deployments should refit and monitor the interaction when judge versions change; this panel cannot establish temporal stability.

Conclusion

Localizing judging inputs can reverse evaluator severity and alter quality gates on identical code. CBC estimates the language-by-evaluator interaction without labels, with a task-level bound allowing dependence within tasks. Removing it improves held-out ranking consistency on DevAI and a separate preference panel, and makes individual-evaluator gates more consistent. The largest interaction exceeds mean repeat-call variation in the sampled slice. Fixed code narrows attribution to the judging setup, but translation differences remain mixed with judge behavior. Panel means are preserved, and improved judging accuracy remains unestablished.

Reproducibility

Results in §5 were recomputed from saved matrices at the reported precision; the artifact documents rare numerical rank-tie differences. It includes preference scores, the corrected human-anchor audit, and the 1,440-call MetaGPT rerun with its prespecified decision file, model routes, and decoding metadata. Human labels come from the public MIT-licensed human_as_a_judge release. The original codebase records Python 3.11, a Poetry lockfile, and

DevAI implementation version 0.1.5. Missing original hardware and provider snapshots prevent exact provider-level replay. Re-run metadata cover 1,000 of 1,440 saved score records; saved-score recomputation remains available. AI-assisted tools were used for wording edits and to assist with implementation of analysis and figure-generation code. The authors specified and verified the design, methodology, calculations, and claims, and remain responsible for the work. Further disclosure and available prompts accompany the scripts, score matrices, and replication records.

Ethics

No new human subjects were recruited; the human labels are public. The correction has not been validated for languages outside the panel. Because a shared language shift survives it, corrected rankings should not be treated as ground truth without an external anchor. The DevAI release is MIT licensed (MIT License, copyright metauto.ai 2024); the local implementation is version 0.1.5 as recorded in `pyproject.toml`.

References

- [1] Anna Bavaresco, Raffaella Bernardi, Leonardo Bertolazzi, Desmond Elliott, Raquel Fernández, Albert Gatt, Esam Ghaleb, Mario Giulianelli, Michael Hanna, Alexander Koller, Andre Martins, Philipp Mondorf, Vera Neplenbroek, Sandro Pezzelle, Barbara Plank, David Schlangen, Alessandro Suglia, Aditya K Surikuchi, Ece Takmaz, and Alberto Testoni. 2025. LLMs instead of Human Judges? A Large Scale Empirical Study across 20 NLP Evaluation Tasks. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*. Association for Computational Linguistics, Vienna, Austria, 238–255. doi:10.18653/v1/2025.acl-short.20
- [2] Benjamin M Bolstad, Rafael A Irizarry, Magnus Astrand, and Terence P Speed. 2003. A Comparison of Normalization Methods for High Density Oligonucleotide Array Data Based on Variance and Bias. *Bioinformatics* 19, 2 (2003), 185–193. doi:10.1093/bioinformatics/19.2.185
- [3] A. P. Dawid and A. M. Skene. 1979. Maximum Likelihood Estimation of Observer Error-Rates Using the EM Algorithm. *Journal of the Royal Statistical Society Series C: Applied Statistics* 28, 1 (1979), 20–28. doi:10.2307/2346806
- [4] Sumanth Doddapaneni, Mohammed Safi Ur Rahman Khan, Dilip Venkatesh, Raj Dabre, Anoop Kunchukuttan, and Mitesh M Khapra. 2025. Cross-Lingual Auto Evaluation for Assessing Multilingual LLMs. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Vienna, Austria, 29297–29329. doi:10.18653/v1/2025.acl-long.1419
- [5] Ora Nova Fandina, Eitan Farchi, Shmulik Froimovich, Rami Katan, Alice Podolsky, Orna Raz, and Avi Ziv. 2025. Automated Validation of LLM-based Evaluators for Software Engineering Artifacts. arXiv preprint. arXiv:2508.02827 [cs.SE] doi:10.48550/arXiv.2508.02827
- [6] Fangxiaoyu Feng, Yinfei Yang, Daniel Cer, Naveen Arivazhagan, and Wei Wang. 2022. Language-agnostic BERT Sentence Embedding. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Dublin, Ireland, 878–891. doi:10.18653/v1/2022.acl-long.62
- [7] Xiyan Fu and Wei Liu. 2025. How Reliable is Multilingual LLM-as-a-Judge?. In *Findings of the Association for Computational Linguistics: EMNLP 2025*. Association for Computational Linguistics, Suzhou, China, 11040–11053. doi:10.18653/v1/2025.findings-emnlp.587
- [8] Srishiti Gureja, Lester James V. Miranda, Shayekh Bin Islam, Rishabh Maheshwary, Drishti Sharma, Gusti Winata, Nathan Lambert, Sebastian Ruder, Sara Hooker, and Marzieh Fadaee. 2025. M-RewardBench: Evaluating Reward Models in Multilingual Settings. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Vienna, Austria, 43–58. doi:10.18653/v1/2025.acl-long.3
- [9] Rishav Hada, Varun Gumma, Adrian de Wynter, Harshita Diddee, Mohamed Ahmed, Monojit Choudhury, Kalika Bali, and Sunayana Sitaram. 2024. Are Large Language Model-based Evaluators the Solution to Scaling Up Multilingual Evaluation?. In *Findings of the Association for Computational Linguistics: EACL 2024*. Association for Computational Linguistics, St. Julian's, Malta, 1051–1070. doi:10.18653/v1/2024.findings-eacl.71
- [10] Sirui Hong, Mingchen Zhuge, Jonathan Chen, Xiaowu Zheng, Yuheng Cheng, Jinlin Wang, Ceyao Zhang, Zili Wang, Steven Ka Shing Yau, Zijuan Lin, Liyang Zhou, Chenyu Ran, Lingfeng Xiao, Chenglin Wu, and Jürgen Schmidhuber. 2024. MetaGPT: Meta Programming for a Multi-Agent Collaborative Framework. In *The Twelfth International Conference on Learning Representations*. <https://openreview.net/forum?id=VtmBAGCN7o>
- [11] W Evan Johnson, Cheng Li, and Ariel Rabinovic. 2007. Adjusting Batch Effects in Microarray Expression Data Using Empirical Bayes Methods. *Biostatistics* 8, 1 (2007), 118–127. doi:10.1093/biostatistics/kxj037
- [12] Shreyas KC. 2026. BabelJudge: Measuring LLM-as-a-Judge Reliability Across Languages and Agent Trajectories. arXiv preprint. arXiv:2606.22329 [cs.CL] doi:10.48550/arXiv.2606.22329
- [13] Chungpa Lee, Thomas Zeng, Jongwon Jeong, Jy-Yong Sohn, and Kangwook Lee. 2026. How to Correctly Report LLM-as-a-Judge Evaluations. In *Proceedings of the 43rd International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 306)*. PMLR, Seoul, South Korea, 65231–65252. doi:10.48550/arXiv.2511.21140
- [14] Haitao Li, Junjie Chen, Qingyao Ai, Zhumin Chu, Yujia Zhou, Qian Dong, and Yiqun Liu. 2025. CalibraEval: Calibrating Prediction Distribution to Mitigate Selection Bias in LLMs-as-Judges. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Vienna, Austria, 16537–16552. doi:10.18653/v1/2025.acl-long.808
- [15] Yanran Li. 2026. Calibrate, Don't Curate: Label-Efficient Estimation from Noisy LLM Judges. arXiv:2605.09702 doi:10.48550/arXiv.2605.09702
- [16] Alhasan Mahmood, Samir Abdaljalil, and Hasan Kurban. 2026. Multilingual Prompt Localization for Agent-as-a-Judge: Language and Backbone Sensitivity in Requirement-Level Evaluation. arXiv:2604.04532 doi:10.48550/arXiv.2604.04532
- [17] Henry Scheffé. 1959. *The Analysis of Variance*. John Wiley & Sons, New York, NY.
- [18] Shayle R Searle. 1971. *Linear Models*. John Wiley & Sons, New York, NY.
- [19] Ivaxi Sheth, Zeno Jonke, Amin Mantrach, and Saab Mansour. 2026. Cross-Lingual LLM-Judge Transfer via Evaluation Decomposition. arXiv:2603.18557 doi:10.48550/arXiv.2603.18557
- [20] Guijin Son, Dongkeun Yoon, Juyoung Suk, Javier Aula-Blasco, Mano Aslan, Vu Trong Kim, Shayekh Bin Islam, Jaume Prats-Cristià, Lucia Tormo-Bañuelos, and Seungone Kim. 2024. MM-Eval: A Multilingual Meta-Evaluation Benchmark for LLM-as-a-Judge and Reward Models. arXiv:2410.17578 doi:10.48550/arXiv.2410.17578
- [21] Rickard Stureborg, Dimitris Alikaniotis, and Yoshi Suhara. 2024. Large Language Models are Inconsistent and Biased Evaluators. arXiv preprint. arXiv:2405.01724 [cs.CL] doi:10.48550/arXiv.2405.01724
- [22] Hiroyasu Usami, Keisuke Hara, Ayato Tsuboi, and Naohiko Matsuda. 2026. LLM Judges Have Dark Current: A Psychometric Datasheet for LLM-as-a-Judge Evaluation. arXiv preprint. arXiv:2606.15610 doi:10.48550/arXiv.2606.15610
- [23] Mingyuan Xu, Xinzi Tan, Jiawei Wu, and Doudou Zhou. 2026. A Judge-Aware Ranking Framework for Evaluating Large Language Models without Ground Truth. arXiv:2601.21817 doi:10.48550/arXiv.2601.21817
- [24] Shaojie Yin. 2026. Does the Judge Prefer English? Evaluating Language-Switching Invariance in LLM-as-a-Judge. arXiv preprint. arXiv:2606.14278 [cs.CL] doi:10.48550/arXiv.2606.14278
- [25] Shibo Yu, Yingzhou Wang, Yan Chen, Guodong Li, and Jin-Hong Du. 2026. Heterogeneous Judge-Aware Ranking with Sensitivity, Disagreement, and Confidence. arXiv:2605.05073 doi:10.48550/arXiv.2605.05073
- [26] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems*, Vol. 36. Curran Associates, Inc., Red Hook, NY, 46595–46623. doi:10.52202/075280-2020
- [27] Ziyi Zhu, Olivier Tieleman, Alexey Bukhtiyarov, and Jinghong Chen. 2026. Cyclic-Judge: Mitigating Judge Bias Efficiently in LLM-based Evaluation. arXiv preprint. arXiv:2603.01865 doi:10.48550/arXiv.2603.01865
- [28] Mingchen Zhuge, Changsheng Zhao, Dylan R. Ashley, Wenyi Wang, Dmitrii Khizbullin, Yunyang Xiong, Zechun Liu, Ernie Chang, Raghuraman Krishnamoorthi, Yuandong Tian, Yangyang Shi, Vikas Chandra, and Jürgen Schmidhuber. 2025. Agent-as-a-Judge: Evaluate Agents with Agents. In *Proceedings of the 42nd International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 267)*. PMLR, Vancouver, Canada, 80569–80611. doi:10.48550/arXiv.2410.10934